



B³/±° / Boba

Multi-tracker meta-search for qBittorrent – self-hosted, containerised, private-tracker-aware.

[Quick start](#) · [Tokens & keys](#) · [Documentation](#) · [Architecture](#) · [Testing](#) · [Contributing](#)

python tests 585 passing frontend tests 182 passing plugins 48 merge service FastAPI :7187
ci auto (syntax / unit / integration / nightly / security)
scanners snyk | sonar | bandit | ruff | semgrep | trivy | gitleaks | pip-audit license Apache 2.0

Features

- **Merge Search Service** – FastAPI service (: 7187) that fans out across 40+ trackers, deduplicates results, streams via SSE.
 - **Real-time results** – result_found events arrive as each tracker completes, no blocking.
 - **Private-tracker bridge** – Authenticated downloads via webui-bridge.py for RuTracker, Kinozal, NNM-Club, IPTorrents.
 - **Freeleech-only IPTorrents** – Automation never costs ratio (see [constitution VIII](#)).
 - **Opt-in quality stack** – SonarQube + Snyk + Semgrep + Trivy + Gitleaks + bandit + pip-audit behind docker-compose.quality.yml.
 - **Opt-in observability** – Prometheus + Grafana dashboards behind the same profile system.
 - **Hook system** – Register scripts via POST /api/v1/hooks for search / download events.
 - **Angular 21 SPA dashboard** – dark-themed, signals-based, per-tracker status chips with CAPTCHA re-login, virtual-scroll-ready sort.
 - **PWA-ready** – favicon + launcher icons + web manifest ship from frontend/public/.
 - **Container-first** – rootless Podman or Docker, single ./start.sh boot.
-

Quick start

```
git clone https://github.com/milos85vasic/qBitTorrent.git
cd qBitTorrent

# Optional: configure tracker credentials + tokens
cp .env.example .env
$EDITOR .env          # see "Tokens & API keys" below

./setup.sh             # one-time
./start.sh             # Angular build + containers + plugin sync
```

Then visit:

URL	Purpose	Auth
http://localhost:7187/	Merge-search dashboard (SPA)	â€”
http://localhost:7187/docs	FastAPI Swagger UI	â€”
http://localhost:7187/openapi.json	OpenAPI schema	â€”
http://localhost:7186/	qBittorrent WebUI proxy	admin / admin
http://localhost:7185/	qBittorrent internal (container)	â€”
http://localhost:7188/	WebUI bridge (private-tracker downloads)	â€”

Tokens & API keys

â”€ **Configure any credentials you need in `.env` before running `./start.sh`.** The single source of truth is [docs/TOKENS_AND_KEYS.md](#) â€” which variable is mandatory, which is optional, and **where to register** for each one.

At a glance

Category	Mandatory	Optional	Documentation
qBittorrent WebUI	â€”	WEBUI_USERNAME, WEBUI_PASSWORD	Â§1
Private trackers	per-tracker	â€”	Â§2
Public tracker uplift	â€”	JACKETT_API_KEY, â€”	Â§3
Metadata enrichment	â€”	TMDB_API_KEY, TVDB_API_KEY, â€”	Â§4
Security scanning	â€”	SNYK_TOKEN, SONAR_TOKEN, â€”	Â§5
Observability	â€”	GRAFANA_USER, GRAFANA_PASSWORD	Â§6
Orchestrator tuning	â€”	ALLOWED_ORIGINS, MAX_CONCURRENT_TRACKERS, â€”	Â§7

Where to register (fast links)

Provider	Signup	Notes
RuTracker	https://rutracker.org/forum/register.php	Username + password
Kinozal	https://kinozal.tv/signup.php	May require invite
NNM-Club	https://nnmclub.to/forum/ucp.php?mode=register	Cookie-based auth
IPTorrents	https://iptorrents.com/	Invite-only; freeleech enforced
Jackett	https://github.com/Jackett/Jackett	Self-hosted indexer
TMDb	https://www.themoviedb.org/signup	v3 auth API key
TVDb	https://thetvdb.com/api-information	Subscription API
MusicBrainz	https://musicbrainz.org/doc/MusicBrainz_API	Free â€” needs UA only
Snyk	https://app.snyk.io/	Account â†’ Auth Token
SonarCloud	https://sonarcloud.io/account/security/	Or run the compose SonarQube
Gitleaks	https://gitleaks.io/	Optional commercial license

Documentation

Everything the platform offers, indexed:

Getting started & operation

- [docs/USER_MANUAL.md](#) â€” end-user walkthrough
- [docs/TOKENS_AND_KEYS.md](#) â€” **â€” credentials / tokens / env vars (mandatory vs optional + registration links)**
- [docs/DESIGN_SYSTEM.md](#) â€” runtime-switchable palette catalogue (8 palettes, dark + light, Darcula default)
- [docs/PLUGINS.md](#) â€” the 48 plugin engines
- [docs/PLUGIN_TROUBLESHOOTING.md](#) â€” what to check when a plugin breaks

Architecture & subsystems

- [docs/architecture/](#) â€” 5 Mermaid diagrams (topology, search lifecycle, plugin execution, bridge, shutdown)
- [docs/DATA_MODEL.md](#) â€” Pydantic schemas & lifecycle (no relational DB)
- [docs/CONCURRENCY.md](#) â€” asyncio semaphore + TTL caches + retry/backoff
- [docs/OBSERVABILITY.md](#) â€” Prometheus + Grafana + OpenTelemetry
- [docs/PERFORMANCE.md](#) â€” benchmark + load + stress test layout
- [docs/api/openapi.json](#) â€” frozen OpenAPI spec (diffed in CI)

Quality & security

- [docs/SECURITY.md](#) â€” threat model + credential storage

- ## Testing

- ## Development

- ## Courses (self-paced, Asciiinema)

- Release / CI**

- # Architecture

Opt-in (docker-compose.quality.yml):

http://:9000	SonarQube	profile: quality
http://:9090	Prometheus	profile: observability

http://:3000	Grafana	profile: observability
(run-once)	Snyk / Semgrep /	
	Trivy / Gitleaks	profile: run-once

Full Mermaid renders in [docs/architecture/](#).

Testing

```
# Non-live-HTTP suites â€” run anywhere, ~10s
python3 -m pytest tests/unit/ tests/e2e/ tests/contract/ --no-cov -q

# Benchmarks â€” dedupe perf + live-HTTP fan-out benchmarks, ~4min
python3 -m pytest tests/benchmark/ --no-cov -q

# Security â€” hits the live merge service, needs stack up
python3 -m pytest tests/security/ --no-cov -q

# Integration â€” same
python3 -m pytest tests/integration/ --no-cov -q -m "not requires_credential"

# Frontend
npx --prefix frontend ng test --watch=false

# Full scanner sweep (non-interactive; skips scanners with missing tokens)
./scripts/scan.sh --all
```

See [docs/TESTING.md](#) for the 30-row test-type catalogue.

Releases

Non-interactive builder at [scripts/build-releases.sh](#) produces artefacts under [releases/<version>/](#):

```
./scripts/build-releases.sh          # all targets, all channels
./scripts/build-releases.sh frontend # one target
./scripts/build-releases.sh --channel release
```

Each artefact ships with SHA256SUMS + BUILD_INFO.json.

Contributing

See [CONTRIBUTING.md](#) and the TDD protocol in [CLAUDE.md](#). PRs must keep the following green:

- Python unit + e2e + contract (py test â€” 585+ tests)
 - Frontend Vitest (ng test â€” 182+ tests)
 - Ruff + bandit + shellcheck (via scripts/scan.sh)
-

License

Apache 2.0 – see [LICENSE](#).